

1.1. V. v. pr. s $\langle \cdot, \cdot \rangle$.

$\forall u, v:$

$$\operatorname{Re}(\langle u, v \rangle) = \frac{1}{4}(\|u+v\|^2 - \|u-v\|^2)$$

$$\operatorname{Im}(\langle u, v \rangle) = \frac{1}{4}(\|u-iv\|^2 - \|u+iv\|^2)$$

$$\frac{1}{4}(\|u+v\|^2 - \|u-v\|^2) = \frac{1}{4}[\langle u+v, u+v \rangle - \langle u-v, u-v \rangle] =$$

$$= \frac{1}{4}[\langle \cancel{u}, \cancel{u} \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle \cancel{v}, \cancel{v} \rangle -$$

$$- \langle \cancel{u}, \cancel{u} \rangle + \langle u, v \rangle + \langle v, u \rangle - \langle \cancel{v}, \cancel{v} \rangle] =$$

$$= \frac{1}{2}(\langle u, v \rangle + \langle v, u \rangle) = \frac{1}{2}(\langle u, v \rangle + \overline{\langle u, v \rangle}) \stackrel{*}{=} \operatorname{Re}(\langle u, v \rangle)$$

* Protože $z + \bar{z} = a+bi + a-bi = 2a = 2 \operatorname{Re}(z)$

$$\frac{1}{4}(\|u-iv\|^2 - \|u+iv\|^2) = \frac{1}{4}[\langle u-iv, u-iv \rangle - \langle u+iv, u+iv \rangle] =$$

$$= \frac{1}{4}[\langle \cancel{u}, \cancel{u} \rangle - \langle u, iv \rangle - \langle iv, u \rangle + \langle \cancel{iv}, \cancel{iv} \rangle -$$

$$- \langle \cancel{u}, \cancel{u} \rangle - \langle u, iv \rangle - \langle iv, u \rangle - \langle \cancel{iv}, \cancel{iv} \rangle] =$$

$$= -\frac{1}{2}(\langle u, iv \rangle + \langle iv, u \rangle) = -\frac{1}{2}(i\langle u, v \rangle + i\overline{\langle u, v \rangle}) =$$

$$= -\frac{1}{2}i(\langle u, v \rangle - \overline{\langle u, v \rangle}) = \operatorname{Im}(\langle u, v \rangle)$$

* Protože $z - \bar{z} = a+bi - a+bi = 2bi = 2i \cdot \operatorname{Im}(z)$